

1. Layer 1: The Physical Layer

This layer deals with the actual physical connection between devices. They just push signals forward.

- **Hub:** connects multiple computers ;broadcasts out to *every* port; high traffic
 - *Types:* Active Hubs (amplify signals), Passive Hubs (just connect cables), Intelligent Hubs (remote monitoring, management, and diagnostic capabilities).
- **Repeater:** regenerate an electrical or optical signal lost due to attenuation
 - *Types:* Analog Repeaters (amplify everything, including noise), Digital Repeaters(removes noise, reconstruct the original signal).
- **Modem (Modulator-Demodulator):** Converts digital signals into analog signals, and vice versa.
 - *Types:* Cable Modems(Coaxial Cable)
 - DSL Modems (Twisted Pair cable)
 - Fiber Modems (OFC)

2. Layer 2: The Data Link Layer

This layer organizes raw bits into "frames" ; gets data from node to node .

Switch: The modern version of the Hub; When data comes in, the switch looks at the destination MAC address and sends the data *only* to that specific device.

- *Types:* Unmanaged Switches (Auto-Negotiation & Full Duplex)
- Managed Switches (Isolated network)
- Multilayer Switch (Local Router)
- **Bridge:** Connects two similar networks. It filters traffic; if data is meant for a computer on the same side of the bridge, it doesn't cross over.
- **Wireless Access Point (WAP):** Allows wireless Wi-Fi devices to connect to a wired network; bridge between the Physical Layer and Data Link Layer.

3. Layer 3: The Network Layer

This layer handles routing data *between* entirely different networks .It uses logical addressing (IP Addresses).

- **Router:** It connects multiple networks together and routes data packets based on their destination IP addresses.
 - *Types:* Core Routers (high speed central line),

- Edge Routers (connecting different networks)

4. Layers 4-7: Transport to Application Layers

Devices operating here look into data reliability and security.

- **Firewall (Works on Layers 3, 4, & 7):** It monitors incoming and outgoing traffic and decides whether to allow or block specific traffic based on protocols
 - *Types:* Network Firewalls (hardware)
 - Stateful Firewalls (Data allowed only if client sends requests)
 - Next-Generation Firewalls (looks into data to block specific app behaviours).
- **Gateway:** Connects different networks and translates the protocols/Data for the receiving network.
- **Load Balancer (Works on Layers 4 & 7):** Distributes incoming network traffic across multiple servers.

L7: Application

PDU: Data

Core Functionality: Network applications interface, protocols (HTTP, FTP, DNS, SMTP, SSH)

Operating Devices & Hardware: Gateways, Next-Gen Firewalls, Reverse Proxies



L6: Presentation

PDU: Data

Core Functionality: Data formatting, translation, encryption, decryption, and compression (SSL/TLS, JPEG, ASCII)

Operating Devices & Hardware: Gateways, Application Firewalls



L5: Session

PDU: Data

Core Functionality: Establishing, managing, synchronizing, and terminating sessions between applications (NetBIOS, RPC)

Operating Devices & Hardware: Gateways



L4: Transport

PDU: Segments / Datagrams

Core Functionality: End-to-end connection control, reliability, error recovery, flow control (TCP, UDP)

Operating Devices & Hardware: Load Balancers, Stateful Firewalls, Gateways

**L3: Network**

PDU: Packets

Core Functionality: Routing, logical addressing (IP addresses), packet forwarding, subnet traffic control (IPv4/IPv6, ICMP)

Operating Devices & Hardware: Routers, Multilayer Switches

**L2: Data Link**

PDU: Frames

Core Functionality: Physical addressing (MAC addresses), switching, frame synchronization, error detection (Ethernet, Wi-Fi)

Operating Devices & Hardware: Switches, Bridges, Wireless Access Points, Network Interface Cards (NIC)

**L1: Physical**

PDU: Bits

Core Functionality: Transmitting raw bitstreams over a physical medium

Devices & Hardware: Hubs, Repeaters, Modems